MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODES, SILICON

TYPE N2146M

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a silicon semiconductor diode. The 1N2146M diode is characterized for use in high current, high speed switching applications and is in accordance with Specification MIL-S-19500, except as otherwise specified herein.

1.2 Maximum ratings.

V_R	I_F	I_R	P 1/	Temperature range	I_f (surge)
Vdc	Adc	mAdc	W	°C.	a
120	1.0	2.0	1.0	-65 to 175	10

1/ Over the ambient temperature range 25° to 175°C., the derating factor is 6.7 mW/°C.

1.3 Primary electrical characteristics.

	V_R	I_s	V_F	C		t_{rr}
	I_R =10μAdc	V_R =50 Vdc	I_F =500 mAdc	$V_{R=0}$	V_R =10 Vdc	I_F = I_R =500mAdc
Min.	120 Vdc	---	---	---	---	---
Max.	---	1.0μAdc	1.1Vdc	45pf	25pf	50 nsec

1.4 Primary quality and reliability assurance requirements. The statistical sampling requirements outlined in the following table are indicative of the requirements of this specification for each of the given reliability levels. The given "estimated maximum failure rate in conservatively designed equipment" is indicative of the achievable failure rates for circuits designed in accord with the "conservative design maximum conditions" specified in 1.5 and the "conservative design circuit failure limits for pertinent characteristics" specified in 1.6.

QUALITY LEVELS (LTPD)				RELIABILITY LEVELS Max. failure rate (λ) during first 1000 hours with 90% confidence.				Estimated maximum failure rate in conservatively designed equipment (%/1000 hrs.)
Group A Subgroups		Group B Subgroups		Steady State Operation Life I_o = 1 Adc v = 120v (peak) f = 60 cps		High Temperature Life (Non-operating) $T_A = (175^{+0}_{-5})°C$		
Major Defect	Minor Defect	Major Defect	Minor Defect	Major Defect	Minor Defect	Major Defect	Minor Defect	
3.0	5.0	10	15	10	15	10	15	0.1

FSC 5961

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1.5 Conservative design maximum conditions. The following "design maximum conditions" are intended as guides for the conservative circuit and system designer. Applications designed to maintain device conditions within these limits may be expected to exhibit an average device failure (see 1.6) rate equal to or below the "estimated maximum failure rate in conservatively designed equipment" shown in the table of 1.4.

V_R	I_F	I_R	P	T_A
100 Vdc	0.5 Adc	1 mAdc	0.5 W	150°C

1.6 Conservative design circuit failure limits for pertinent characteristics. The "Minor" failure limits given in the following table represent reasonable characteristic requirements for conservatively designed circuits. The "Major" failure limits represent characteristic values for which the probability of failure in conservatively designed equipment is relatively high.

Defect Class	Limit	BV I_R = 10μAdc	I_S V_R = 50Vdc	V_F I_F = 500mAdc	C		t_{rr} $I_F = I_R$ = 500 mAdc
					V_R = 0	V_R = 10Vdc	
Minor	Min. Max.	100 Vdc ---	--- 2μAdc	--- 1.4 Vdc	--- 50 pf	--- 25 pf	--- 60 nsec
Major	Min. Max.	80 Vdc ---	--- 4μAdc	--- 2.2 Vdc	--- 90 pf	--- 60 pf	--- 100 nsec

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATION

MILITARY
MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Diodes shall be in accordance with Specification MIL-S-19500 and as specified herein.

3.2 Design and construction. Semiconductor diodes shall be of the design, construction and physical dimensions as shown on figure 1.

3.3 Performance characteristics. Performance characteristics shall be as specified in 4.3 and 4.4.

3.4 Marking. Diodes shall be marked with the "USN" prefix in lieu of the "JAN" prefix. The cathode shall be indicated by a contrasting band. The type number shall include the "M" suffix. Country of origin and manufacturer's identification may be omitted from the device.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Qualification tests. Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. Qualification tests shall consist of the tests specified in 4.3 and 4.4. (Application for qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.1).)

4.2 Acceptance inspection. Acceptance inspection shall consist of the examinations and tests specified in 4.3 and 4.4.

4.2.1 Acceptance procedure. When a second sample is chosen, the total sample shall be that sample associated with an acceptance number of one less than the minimum rejection numbers specified.

4.2.2 Acceptance inspection information. When specified in the contract or order, one copy of the acceptance inspection data pertinent to the inspection lot shall accompany the shipments.

4.2.3 Visual and mechanical examination. Major and minor defects for visual and mechanical examination shall be defined in Standard MIL-STD-105.

4.2.4 The lot of devices may be rejected if the observed number of defectives exceeds five and the proportion defective exceeds the specified LTPD or λ for any test subgroup.

4.3 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.4 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.4.1 Destructive tests. The tests in subgroup 4 and 5 of Group B inspection are considered destructi

4.4.2 Salt atmosphere. The device shall be examined for destructive corrosion and illegible marking.

5. PREPARATION FOR DELIVERY

5.1 See Specification MIL-S-19500.

6. NOTES

6.1 The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington 25, D. C., and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification". (Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pa.)

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Preparing activity:
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Table I - Group A inspection

Examination or test	CONDITIONS		SAMPLING		Symbol	LIMITS				UNITS
	MIL-STD-750 Method	Specific Conditions	Minor LTPD (λ) %	Major LTPD (λ) %		Minor		Major		
						Min.	Max.	Min.	Max.	
Subgroup 1										
Visual and mechanical examination	2071	----	10	7	----	---	---	---	---	----
Subgroup 2										
Saturation current	4076	V_R = 50 Vdc			I_S	---	1.0	---	2.0	uAdc
Forward current	4011	I_F = 0.5 Adc			V_F	---	1.1	---	2.2	Vdc
Breakdown voltage	4021	I_R = 10 uAdc			BV	120	---	100	---	Vdc
Capacitance	4001	V_R = 0 f = 100 kc	5	3	C	---	45	---	90	pf
Capacitance	4001	V_R = 10 Vdc f = 100 kc			C	---	25	---	40	pf
Reverse recovery time	4031 Condition B	I_F=I_R=500 mAdc RC = 5 nsec			t_{rr}	---	50	---	75	nsec

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Table II - Group B inspection.

Examination or test	CONDITIONS		SAMPLING		Symbol	LIMITS				UNITS
	MIL-STD-750 Method	Specific Conditions	Minor LTPD (λ) %	Major LTPD (λ) %		Minor		Major		
						Min.	Max.	Min.	Max.	
Subgroup 1 Physical dimensions	2066	----	10	5	---	---	---	---	---	---
Subgroup 2 Soldering heat	2031	---			---	---	---	---	---	---
Temperature cycling	1051 Cond. B	10 cycles, $T_{(high)}$=175°C.	15	10	---	---	---	---	---	---
Thermal shock	1056 Cond. A	----			---	---	---	---	---	---
Moisture resistance	1021	----			---	---	---	---	---	---
Subgroup 3 Shock	2016	5 blows, X1, Y1 and Y2, 1500G, 0.5 msec			---	---	---	---	---	---
Vibration, variable frequency	2056	----	15	10	---	---	---	---	---	---
Vibration fatigue	2046	----			---	---	---	---	---	---
Constant acceleration	2006	10,000G, X1, Y1, and Y2			---	---	---	---	---	---
Subgroup 4 Lead fatigue	2036 Cond. E	----	15	10	---	---	---	---	---	---
Subgroup 5 Salt atmosphere	1041	----	15	10	---	---	---	---	---	---

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Table II - Group B inspection (Con't)

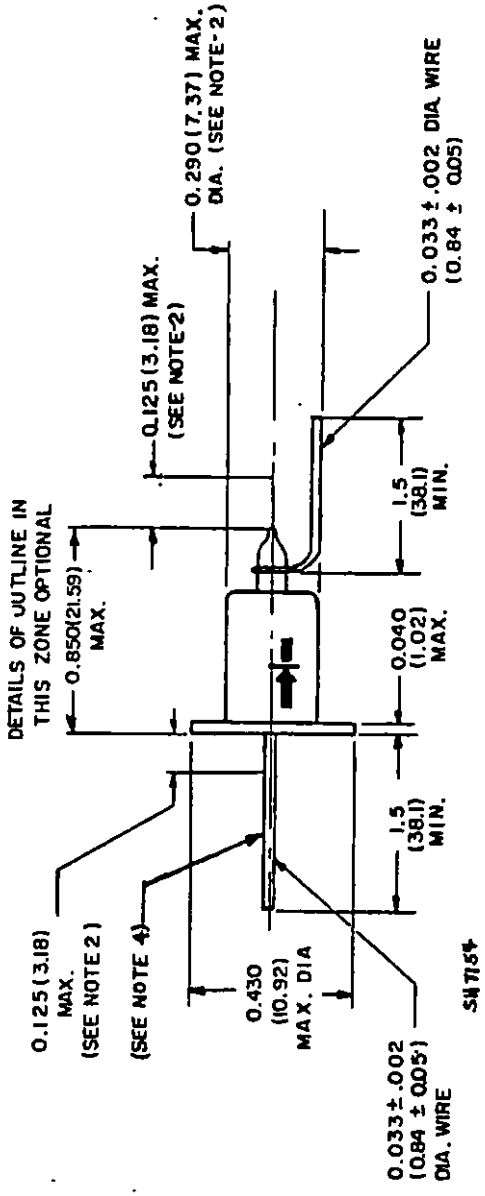
Examination or test	CONDITIONS		SAMPLING		Symbol	LIMITS				UNITS
	MIL-STD-750 Method	Specific Conditions	Minor LTPD (λ) %	Major LTPD (λ) %		Minor		Major		
						Min.	Max.	Min.	Max.	
Subgroup 6										
Maximum reverse current	4016	I_R = 2.0mAdc, 10 ± 1 min.								
Surge current	4066	I_o = 1 Adc $I_{f(surge)}$=10a, t=1/120sec, ten surges, 1 surge/ min.	15	10	---	---	---	---	---	---
Subgroup 7										
Thermal resistance	3151	----	15	10	θ_J-A	---	150	---	---	°C/mW
Subgroup 8										
Saturation current	4076	V_R = 50 Vdc T_A = 150° ± 5°C.			I_S	---	2	---	4	mAdc
Dew point	1066	$T_{(high)}$=25°±5°C. $T_{(low)}$=-40°±5°C.	15	10	---	---	---	---	---	---
Saturation current	4076	V_R = 50 Vdc			I_S	---	5	---	10	uAdc
Subgroup 9										
High-temperature life (nonoperating)	1031	T_A=175°$^{+0°C.}_{-5°C.}$	15	10	---	---	---	---	---	---
Subgroup 10										
Steady state operation life	1026	I_0 = 1 Adc v =120 v (peak) f = 60 cps	15	10						

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Table II - Group B inspection (Con't)

Examination or test	CONDITIONS		SAMPLING		Symbol	LIMITS				UNITS
	MIL-STD-750 Method	Specific Conditions	Minor LTPD (λ) %	Major LTPD (λ) %		Minor		Major		
						Min.	Max.	Min.	Max.	
End points (subgroups 2, 3, 5, 6, 7 and 8):										
Breakdown voltage	4021	I_R = 10 uAdc	--	--	BV	120	---	100	---	Vdc
Forward voltage	4011	I_F = 0.5 Adc	--	--	V_F	---	1.1	---	1.2	Vdc
Saturation current	4076	V_R = 50 Vdc	--	--	I_S	---	1.0	---	2.0	uAdc
End points (subgroups 9 and 10):										
Breakdown voltage	4021	I_R = 10 uAdc	--	--	BV	120	---	100	---	Vdc
Forward voltage	4011	I_F = 0.5 Adc	--	--	V_F	---	1.1	---	2.2	Vdc
Saturation current	4076	V_R = 50 Vdc	--	--	I_S	---	2.0	---	4.2	Vdc
Capacitance	4001	V_R = 10 Vdc	--	--	C	---	25	---	40	pf

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NOTES:

1. Leads may be located on any parallel axis within the 0.290 diameter.
2. Lead diameter uncontrolled in this area to allow for paint, solder, run-up, and so forth.
3. All dimensions in inches. Dimensions in parenthesis are in millimeters.
4. This lead is electrically connected to case.

Figure 1 - Outline dimensions of diode type 1N2146M.

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